Hands on metabarcoding - beta diversity

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Beta diversity

For the beta diversity analysis, we will use use the **Vegan** package, which means that we will be working mostly with **matrices** and **list**, and we will make **plots in base R**.

Recall that we made a matrix in a previous hands on session, **ASVs_wide_t**, this will be our starting point.

Calculate non metric Multidimensional Scaling (mMDS) (1)

Using the **ASVs_wide_t** matrix, calculate nMDS, using **metaMDS()** function with default parameters (which include the Bray-Curtis dissimilarity). Call the new nMDS object **ASVs_nMDS**:

```
----
```

Calculate non metric Multidimensional Scaling (mMDS) (2)

Solution:

```r
ASVs_nMDS <- metaMDS(ASVs_wide_t)
```

```
## Square root transformation
## Wisconsin double standardization
## Run 0 stress 0.1539068
## Run 1 stress 0.160366
## Run 2 stress 0.1500365
## ... New best solution
## ... Procrustes: rmse 0.1354972  max resid 0.3549388
## Run 3 stress 0.1504577
## ... Procrustes: rmse 0.09340805  max resid 0.2476451
## Run 4 stress 0.1942333
## Run 5 stress 0.1510932
## Run 6 stress 0.1510937
## Run 7 stress 0.1600158
## Run 8 stress 0.1636527
## Run 9 stress 0.1538919
## Run 10 stress 0.1504576
## ... Procrustes: rmse 0.09340079  max resid 0.2477181
## Run 11 stress 0.1532057
## Run 12 stress 0.1895481
## Run 13 stress 0.1603654
```

Prepare aesthetics in a dedicated data frame (1)

Since we are going to use base R for the plots, we will have some extra work to produce good plots.

Start by making a data frame with all the relevant metadata, which will already include the coloring based on Experiment and Treatment. Call the new object **beta_env**:

```
____ <- metadata_clean %>%
  mutate(Experiment_color = ifelse(Experiment == "Sed002",
                                   qualitative_colors[1],
                                   qualitative_colors[2])) %>%
  mutate(Treatment_color = case_when(Treatment == "CTR" ~ greens[1],
                                     Treatment == "Cd 0.015" ~ greens[2],
                                     Treatment == "Cd 0.15" ~ greens[3],
                                     Treatment == "Cd 1.5" ~ greens[4],
                                     Treatment == "Cd 15" ~ greens[5])) %>%
  mutate(Cd = Concentration)
```

Prepare aesthetics in a dedicated data frame (2)

Solution:

```
beta_env <- metadata_clean %>%
  mutate(Experiment_color = ifelse(Experiment == "Sed002",
                                   qualitative_colors[1],
                                   qualitative_colors[2])) %>%
  mutate(Treatment_color = case_when(Treatment == "CTR" ~ "grey",
                                     Treatment == "Cd 0.015" ~ "#C7E9C0",
                                     Treatment == "Cd 0.15" ~ "#74C476",
                                     Treatment == "Cd 1.5" ~ "#238B45",
                                     Treatment == "Cd 15" ~ "#00441B")) %>%
  mutate(Cd = Concentration)
```

Prepare aesthetics in a dedicated data frame (3)

The environmental information on the data frame must perfectly match with the matrix of species abundance and nMDS. Thus, make sure that beta_env does not have any extra samples:

```
beta_env <- beta_env %>% filter(Sample %in% rownames(seqtab.nochim))
```

Prepare aesthetics in a dedicated data frame (4)

Add explicit row names to the **beta_env** object, to be able to make a connection with the nMDS object later on.

```
# ignore the warning
____(beta_env) <- beta_env$Sample
```

Prepare aesthetics in a dedicated data frame (5)

Solution:

```
# ignore the warning
rownames(beta_env) <- beta_env$Sample
```

Plot nMDS (1)

To plot the nMDS we will need several layers.

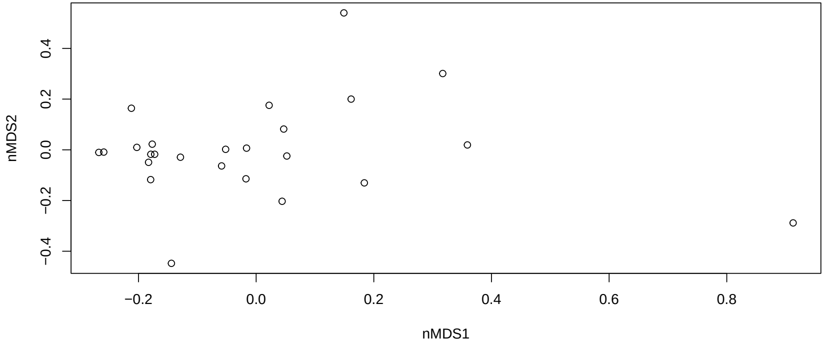
Lets start by the default plot. Use the **points** data from the **ASVs_nMDS** object and set display type to **p**.

```
plot(____$points,
     display = ____,
     type = "p",
     xlab = "nMDS1",
     ylab = "nMDS2")
```

Plot nMDS (2)

Solution:

```
plot(ASVs_nMDS$points, type = "p",
     xlab = "nMDS1", ylab = "nMDS2")
```



Plot nMDS (3)

Add a new layer to edit the points aesthetics.

Use the **points()** function with the **ASVs_nMDS** data. Then, with the function **abline()** add the x and y axis crossing zero.

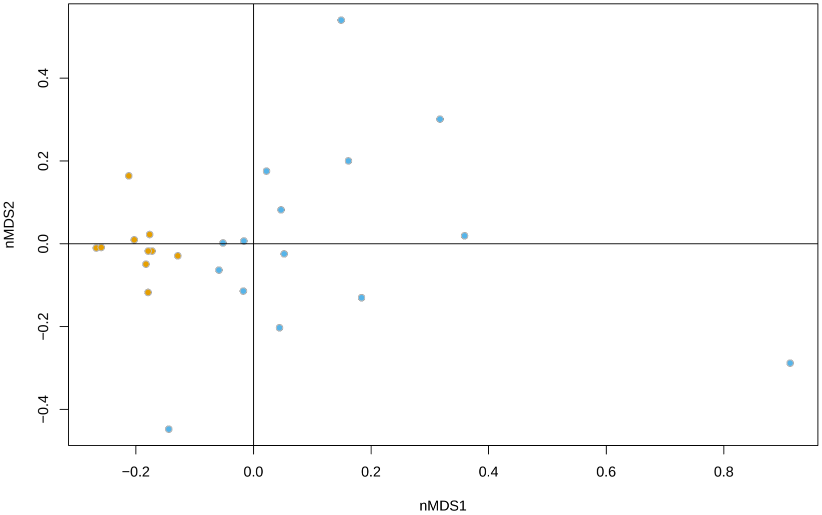
```
plot(ASVs_nMDS$points, type = "p",
     xlab = "nMDS1",ylab = "nMDS2")
____(____,
       bg = beta_env$Experiment_color,
       pch = 21,
       col = "grey",
       cex = 1)
abline(h = ____, v = ____)
```

Plot nMDS (4)

Solution:

```
plot(ASVs_nMDS$points, type = "p",
     xlab = "nMDS1",ylab = "nMDS2")
points(ASVs_nMDS,
       bg = beta_env$Experiment_color,
       pch = 21, col = "grey",
       cex = 1)
abline(h = 0, v = 0)
```

Plot nMDS (5)



Plot nMDS (6)

Add a color legend in the top right position, using the **legend()** function:

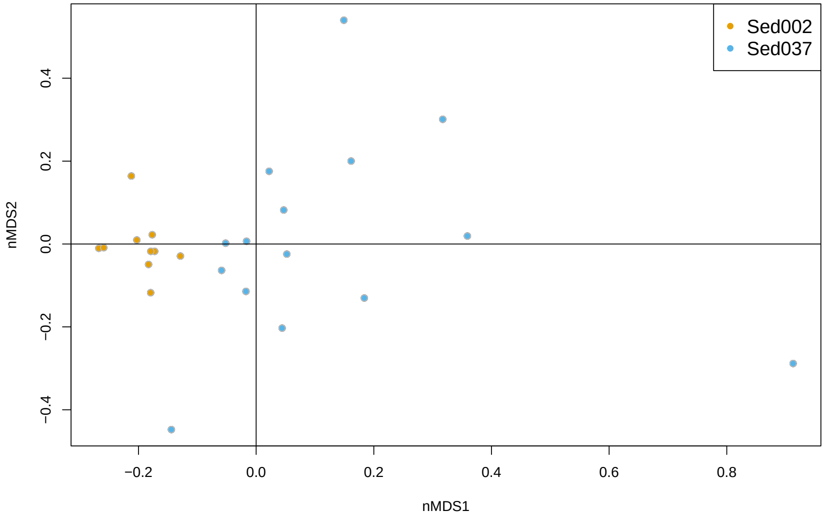
```
plot(ASVs_nMDS$points, type = "p",
     xlab = "nMDS1",ylab = "nMDS2")
points(ASVs_nMDS,
       bg = beta_env$Experiment_color,
       pch = 21, col = "grey",
       cex = 1)
abline(h = 0, v = 0)
____("topright",
       legend=c("Sed002", "Sed037"),
       pch=20, col=qualitative_colors[c(1,2)],cex =1.3)
```

Plot nMDS (7)

Solution:

```
plot(ASVs_nMDS$points, type = "p",
     xlab = "nMDS1",ylab = "nMDS2")
points(ASVs_nMDS,
       bg = beta_env$Experiment_color,
       pch = 21, col = "grey",
       cex = 1)
abline(h = 0, v = 0)
legend("topright", legend=c("Sed002", "Sed037"),
       pch=20, col=qualitative_colors[c(1,2)],cex =1.3)
```

Plot nMDS (8)



Plot nMDS (9)

We can add a vector indicating the direction and magnitude of the Cd concentration variable.

To do so, calculate the vector using **envfit()** function and store in an object called **Cd_fit**.

```
____ <- ____(ASVs_nMDS ~ Cd, beta_env)
```

Plot nMDS (10)

Solution:

```
Cd_fit <- envfit(ASVs_nMDS ~ Cd, beta_env)
```

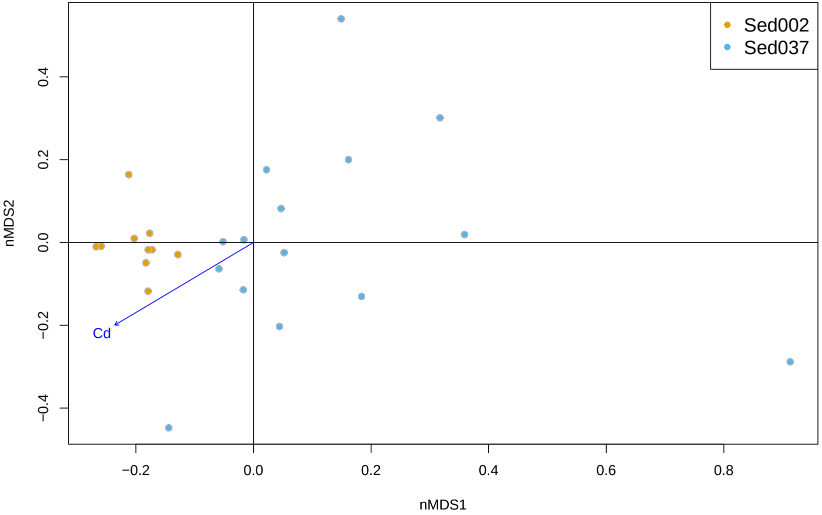
Plot nMDS (11)

Add the vector to the previous plot. To do so, plot the **Cd_fit** directly after the first plot:

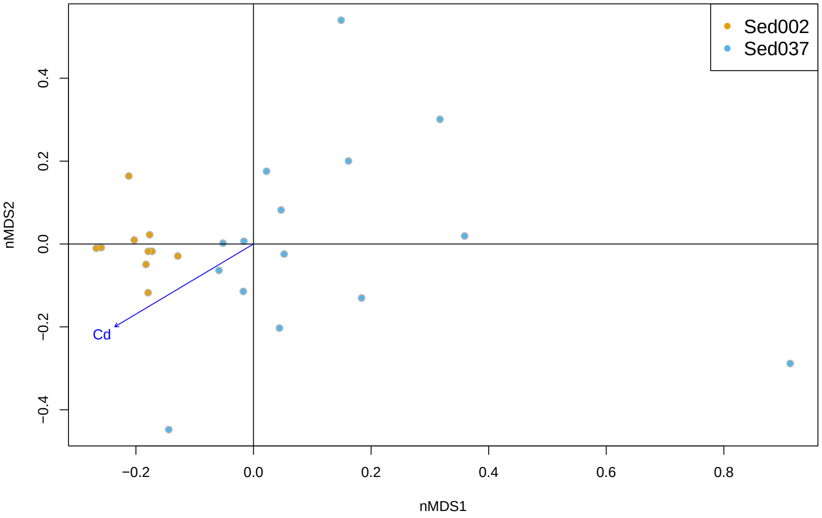
```
plot(ASVs_nMDS$points, type = "p",
     xlab = "nMDS1",ylab = "nMDS2")
points(ASVs_nMDS,
       bg = beta_env$Experiment_color,
       pch = 21, col = "grey",
       cex = 1)
abline(h = 0, v = 0)
plot(____)
legend("topright", legend=c("Sed002", "Sed037"),
       pch=20, col=qualitative_colors[c(1,2)],cex =1.3)
```

Plot nMDS (12)

Solution:



Plot nMDS (13)



Plot nMDS (14)

To exemplify the addition of a factor variable, lets use the Experiment variable.

Use the function **with()** to add an ellipse to the ordination, centered at the centroid of each experiment.

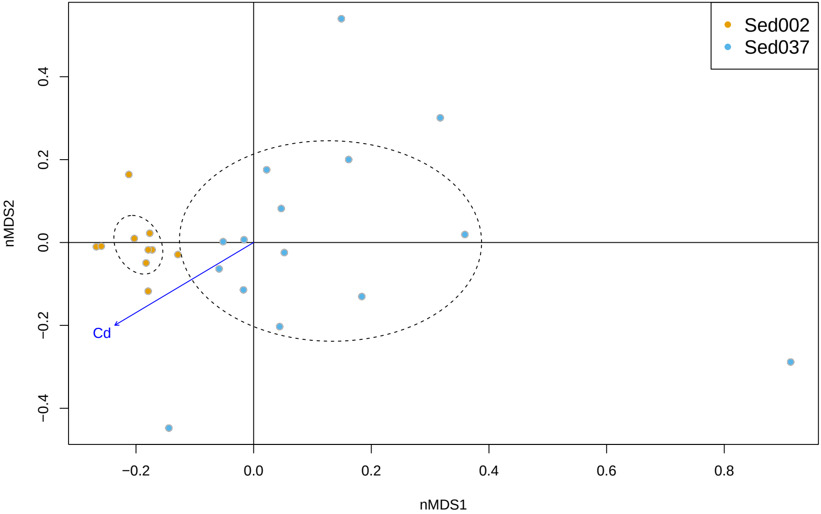
```
plot(ASVs_nMDS$points, type = "p",
     xlab = "nMDS1",ylab = "nMDS2")
points(ASVs_nMDS,
       bg = beta_env$Experiment_color,
       pch = 21, col = "grey",
       cex = 1)
abline(h = 0, v = 0)
plot(Cd_fit)
legend("topright", legend=c("Sed002", "Sed037"),
       pch=20, col=qualitative_colors[c(1,2)],cex = 1.3)
____(beta_env,
     ordiellipse(ASVs_nMDS, Experiment,
                 lty = "dashed",
                 label = FALSE,
                 cex = 1.3))
```

Plot nMDS (15)

Solution:

```
plot(ASVs_nMDS$points, type = "p",
     xlab = "nMDS1",ylab = "nMDS2")
points(ASVs_nMDS,
       bg = beta_env$Experiment_color,
       pch = 21, col = "grey",
       cex = 1)
abline(h = 0, v = 0)
plot(Cd_fit)
legend("topright", legend=c("Sed002", "Sed037"),
       pch=20, col=qualitative_colors[c(1,2)],cex = 1.3)
with(beta_env,
     ordiellipse(ASVs_nMDS, Experiment,
                 lty = "dashed",
                 label = FALSE,
                 cex = 1.3))
```

Plot nMDS (16)

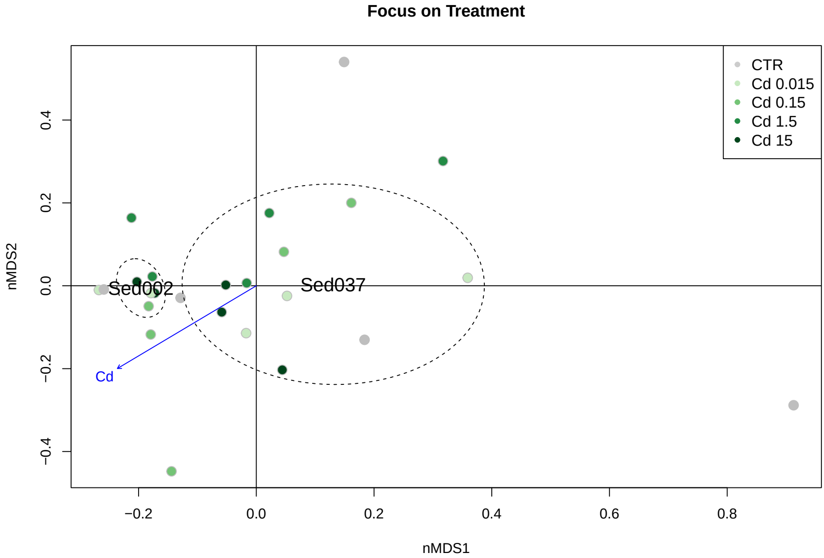


Plot nMDS (17)

We can also use coloring to illustrate the Treatment instead of the Experiment:

```
plot(ASVs_nMDS$points, type = "p",
     xlab = "nMDS1",ylab = "nMDS2", main = "Focus on Treatment")
abline(h = 0, v = 0)
points(ASVs_nMDS,
       bg = beta_env$Treatment_color,
       pch = 21, col = "grey",
       cex = 1)
plot(Cd_fit)
legend("topright", legend = unique(beta_env$Treatment),
       pch=20, col=greens,cex = 1.1)
with(beta_env,
     ordiellipse(ASVs_nMDS, Experiment,
                 lty = "dashed",
                 label = TRUE,
                 cex = 1.3))
```

Plot nMDS (18)



Focus on Treatment
CTR
Cd 0.015
Cd 0.15
Cd 1.5
Cd 15
Sed002
Sed037
Cd
nMDS1
nMDS2
−0.2
0.0
0.2
0.4
0.6
0.8
−0.4

PERMANOVA test (1)

Did the experiment and/or treatment have a significant impact on the community composition?

To test that with the PERMANOVA we need to:

- Test homogeneity of variance of each variable, using **betadisper()**;
- Test PERMANOVA, with appropriate model.

PERMANOVA test (2)

To test homogeneity of variance, we can do permutations of the betadisper() function, which will run multivariate version of the Levene test.

This test can also be used as stand alone to compare variance of two groups.

Use the **betadisper()** function inside the permutations:

```
set.seed(123); permutest(
  ____(
    vegdist(ASV_rarefied),
    group = beta_env$Experiment),
  permutations = 999)
```

PERMANOVA test (2)

Solution:

```r
set.seed(123); permutest(
  betadisper(
    vegdist(ASV_rarefied),
    group = beta_env$Experiment),
  permutations = 999)
```

```
##
## Permutation test for homogeneity of multivariate dispersions
## Permutation: free
## Number of permutations: 999
##
## Response: Distances
##           Df   Sum Sq  Mean Sq      F N.Perm Pr(>F)
## Groups     1 0.040941 0.040941 7.5415    999  0.006 **
## Residuals 23 0.124862 0.005429
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

There were significant differences between both Experiments variability.

PERMANOVA test (3)

Repeat the same analysis for the **Treatment** variable:

```
permutest(
  betadisper(
    vegdist(ASV_rarefied),
    group = beta_env$____),
  permutations = 999)
```

PERMANOVA test (3)

Repeat the same analysis for the **Treatment** variable:

```
set.seed(123); permutest(
  betadisper(
    vegdist(ASV_rarefied),
    group = beta_env$Treatment),
  permutations = 999)
```

```
##
## Permutation test for homogeneity of multivariate dispersions
## Permutation: free
## Number of permutations: 999
##
## Response: Distances
##           Df   Sum Sq   Mean Sq      F N.Perm Pr(>F)
## Groups     4 0.013337 0.0033343 0.7364    999  0.555
## Residuals 20 0.090553 0.0045276
```

There were no significant differences in the variability of both treatments.

PERMANOVA test (4)

Now we will do the PERMANOVA test, but we should be aware that both Experiments do not have an homogeneous variation in community composition.

Use the **adonis2()** function to calculate PERMANOVA, testing only the **Experiment** variable. Use the original species abundance table matrix,

```
set.seed(123); ____(ASVs_wide_t ~ Experiment, data = beta_env)
```

PERMANOVA test (5)

Solution:

```
set.seed(123); adonis2(ASVs_wide_t ~ Experiment, data = beta_env)
```

```
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 999
##
## adonis2(formula = ASVs_wide_t ~ Experiment, data = beta_env)
##          Df SumOfSqs      R2      F Pr(>F)
## Model     1  0.73654 0.23449 7.0454  0.002 **
## Residual 23  2.40447 0.76551
## Total    24  3.14101 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The community composition was **significantly different** between Experiments.

PERMANOVA test (6)

Repeat for the Treatment variable.

```
set.seed(123); adonis2(ASVs_wide_t ~ Treatment, data = beta_env)
```

```
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 999
##
## adonis2(formula = ASVs_wide_t ~ Treatment, data = beta_env)
##          Df SumOfSqs      R2      F Pr(>F)
## Model     4  0.52822 0.16817 1.0108  0.394
## Residual 20  2.61279 0.83183
## Total    24  3.14101 1.00000
```

The community composition did **not change significantly** between Treatments.

PERMANOVA test (7)

However, it seems unfair to compare Treatments mixed with Experiments, because the Experiments were significantly different in composition and variability.

Adapt the formula to compare **Treatments** within each **Experiment**:

```
set.seed(123); adonis2(ASV_rarefied ~ Treatment*____, data = beta_env)
```

PERMANOVA test (8)

Solution:

```
set.seed(123); adonis2(ASV_rarefied ~ Treatment*Experiment, data = beta_env)
```

```
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 999
##
## adonis2(formula = ASV_rarefied ~ Treatment * Experiment, data = beta_env)
##          Df SumOfSqs      R2      F Pr(>F)
## Model     9   1.6957 0.48781 1.5874  0.017 *
## Residual 15   1.7804 0.51219
## Total    24   3.4762 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

This last test confirms that the experiments do have significantly different community composition, but the Treatments do not.